

Highlight of Skills

IT skills: Programming languages (R, Python, SQL), QGIS, Tableau, Microsoft Office software

Languages: English, Cantonese, Mandarin

Experienced in working as a team with high reliability and self-motivation

Strong problem-solving and critical thinking skills, attentive to details

Education

Sep 2025 – **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada

Apr 2026 (Expected) Courses: Applied Multivariate Statistics, Special Topics in Statistical Data Science, Survival Analysis

Sep 2021 – **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada

cGPA: 3.94/4.00

Courses: Applied Statistical Methods, Spatial Data Analysis, Time Series Analysis, Surveys, Sampling and Observational Data, Statistical Consultation, Communication, and Collaboration.

Academic awards: Samuel Beatty In-course Scholarship (2025), Dean's List (2021–2025), University of Toronto Scholar (2021)

Statistical Professional and Research Experiences

Sep 2025 – **Statistical Neuroimaging Researcher**, *University of Toronto, Department of Statistical Sciences*

- Current
- Conducted research on statistical inference for the genetic heritability of intermodal coupling using multi-modal brain MRI data.
 - Explored adaptive clustering approaches and spatiotemporal correlation modeling techniques (e.g. CEIDR, CLEAN) to control Type I error rates and increase statistical power effectively, strengthening research critical thinking skills.
 - Performed large-scale simulation studies in R using parallel computing to evaluate and validate proposed intermodel-correlation statistical method, developing expertise in statistical computing.
 - Summarized research findings in report and presented during weekly meetings, enhancing statistical communication skills.

Sep 2025 – **R-Based Learning Designer**, *University of Toronto Human Biology Program*

- Current
- Developed 4 testable R coding assignments for junior undergraduate life science students with minimal statistical background in student learning platform MarkUs, improving students' accessibility in statistical learning as well as pedagogical effectiveness.
 - Collaborated with faculty member to develop 2 LearnR modules content for senior undergraduate life science students, filling content gaps and adjusting pacing to improve students' comprehension and engagement in statistical learning.
 - Applied strong attention to detail when reviewing module content and coding exercises, improving instructional quality and usability of learning resources.

Aug 2025 – **LearnR Module Tester**, *University of Toronto Department of Statistical Sciences*

- Current
- Conducted testing on 4 R-based learning modules designed for junior undergraduate statistics student, identifying bugs, errors, and gaps while evaluating module content material and progression, improving the overall quality of learning resources.
 - Collaborated and communicated progress with module developer to address issues and revise updated modules, refining teamwork and technical communication skills.
 - Provided detailed written feedback on codes accuracy and learning experience from the perspective of an undergraduate statistics student, contributing to improvements in instructional design and strengthened critical thinking skills.

- May 2024 **Regression Analysis Researcher, University of Toronto Department of Statistical Sciences**
 – Current
- Conducted research on project “Aspects of Robust Regression Analysis”, supported by University of Toronto Excellence Award: Natural Sciences and Engineering and University of Toronto DoSS Summer Undergraduate Research Award, leading to a preprint paper and a research day presentation.
 - Developed and evaluated 4 approaches for estimating parameters in student-t regression using simulated data under high-dimensional and outlier regression scenarios, which led to a comprehensive prediction performance analysis of estimators and contributed insights for improving traditional model accuracy.
 - Implemented and troubleshoot various optimization algorithms to address non-convergence issues and enhanced computational efficiency, enhancing proficiency for statistical computing and simulation through utilizing software R and Python.
 - Presented research findings in report format and visualizations, strengthening skills in clear communication of complex statistical methodologies and results.
- Sep 2024 – **Machine Learning Researcher, University of Toronto Department of Statistical Sciences**
 Apr 2025
- Conducted research on project “Fair machine learning with biased label”, leading to a research day presentation.
 - Reviewed statistical literature on fairness in machine learning and machine learning embedding algorithms, strengthening research synthesis skills and producing a comprehensive summary of key methods and challenges.
 - Simulated and evaluated embedding-building algorithms in R using known pairs to demonstrate the effectiveness of existing techniques in capturing relationship between 6000+ medical concepts and uncovering key predictors for downstream modeling.
 - Utilized supervised machine learning models (logistic regression and random forest) for depression feature selection, developing practical modeling expertise and providing insights into demographic group bias in prediction model.
- Sep 2024 – **Tableau Data Visualization Analyst, University of Toronto Human Biology Program**
 Apr 2025
- Analyzed student data from learning platform Quercus and course grades using Excel to identify trends and correlations that inform evidence-based teaching strategies, translating quantitative analyses into actionable insights for course improvement.
 - Created 100+ data visualizations in Tableau to summarize results and communicated complex findings clearly to faculty members during regular progress meetings, refining critical thinking and statistical communication skills.
 - Collaborated with peers to troubleshoot data challenges and refine analytical approaches, strengthening problem-solving and teamwork skills.
- Jun 2024 – **Psychiatric Education Quantitative Researcher, Centre for Addiction and Mental Health**
 Apr 2025
- Conducted research on project “Investigating the efficacy of a novel approach to opioid overdose training using virtual reality”, supported by University of Toronto Institute of Medical Science Department of Psychiatry Summer Undergraduate Research Program Award, resulting in a paper publication and poster presentation.
 - Performed statistical analyses in R to examine the relationship between cybersickness and degree of physical movement in immersive VR simulation-based psychiatric medical educational programs, revealing that higher physical activity in VR training correlates with increased nausea.
 - Summarized descriptive statistics and identified trends in participant tolerability and symptom severity across different training scenarios, contributing valuable insights for future research on minimizing cybersickness in VR environments.
 - Provided data-driven recommendations for the design of VR training programs that prioritize user comfort, emphasizing the importance of movement considerations while maintaining educational efficacy.
- Sep 2023 – **Agent-Based Modeling Researcher, University of Toronto COBWEB lab**
 Apr 2024
- Initiated research on project “An Agent-Based Simulation Approach to Investigate the Effect of Decreasing Birth Rates and the Efficacy of Potential Solutions”, which led a poster presentation.
 - Conducted comprehensive literature reviews and prepared an annotated bibliography, synthesizing findings from multidisciplinary sources and enhancing critical thinking and organization skills.
 - Boosted self-learning skills by mastering the simulation software COBWEB, utilizing its various functionalities to simulate population dynamics and policy scenarios.
 - Performed time-series and regression analyses in Python to assess the impact of policy interventions on population dynamics and birth rates trends, providing insights for short-term and long-term planning.
- May 2023 **Data Analyst Intern, Canadian Urban Institute**
 – Aug 2023
- Developed multivariate linear regression models and multinomial logistic regression model in R to analyze nationwide household and business-level carbon emissions, and social connection demographical patterns, strengthening applied statistical modeling capability.
 - Created an interactive situational tool to demonstrate the effects of urbanization and housing composition on greenhouse gas emissions, demonstrating skills and data visualization, tool development and stakeholder communication.
 - Utilized spatial analysis and geospatial visualization in QGIS to produce 300+ detailed maps, highlighting urbanization and housing impacts on greenhouse gas emissions.
 - Synthesized statistical findings into policy-relevant urban planning insights during weekly meetings, demonstrating ability to translate technical results into actionable recommendations for decision-makers.

- Oct 2022 – **Computational Social Science Researcher**, *University of Toronto Population Well-being Lab*
- May 2023
- Conducted research on project “Life Satisfaction Outcomes for Domain Satisfied Individuals: Clustering Domain Satisfaction”, resulting in three conference presentations and Eleventh Annual Canadian Statistics Student Conference Undergraduate Level Poster First Prize.
 - Co-drafted the research plan, demonstrating competency in strategic planning by outlining clear objectives, methodologies, and timelines.
 - Performed data analysis using unsupervised clustering methods, including multivariate gaussian mixture model, k-means and hierarchical clustering, improving technical proficiency and R programming skills.
 - Evaluated effectiveness and appropriateness of different statistical methods, created detailed tip-sheets on algorithms such as expectation-maximization and the elbow method, enhancing understanding and application of these algorithms.
 - Developed strong statistical communication skills, effectively designed and delivered presentations that discussed the suitability and implications of different clustering methods used, honing knowledge translation abilities.
- Sep 2022 – **Statistical Consultant**, *University of Toronto Data Sciences Café*
- Dec 2023
- Contributed to the development of a supportive and collaborative consulting community, fostering an environment where interdisciplinary research and statistical analysis could thrive.
 - Acquired expertise in recommending appropriate statistical methodologies to guest speakers based on their specific needs and research objectives, cultivating skills in analyzing multidisciplinary datasets and tailoring analysis to diverse fields.
 - Identified potential biases and limitations in proposed research methods, providing actionable insights for improvement in study method design.
 - Prepared and presented monthly reports on various statistical topics, demonstrating statistical approaches and coding methods in an accessible manner to enhance statistical literacy among non-statistician researchers.

Other Experiences

- Sep 2023 – **Teaching Assistant and Learning Support**, *University of Toronto Department of Statistical Sciences and Data Sciences Institute*
- Current
- DoSS Courses: STA130, STA238, STA304, Statistics Aid Centre, Florence Nightingale Day (2023, 2024)
- DSI Modules: SQL, Applying Statistical Concepts, Sampling, Visualization
- Prepared and delivered tutorials and office hours for statistics courses, strengthening programming (R, Python, SQL) skills and facilitated learning by translating technical content into digestible explanations, enhancing students’ statistical reasoning and analytical skills.
 - Assisted undergraduate students at the Statistic Aid Centre, addressing questions across a wide range of statistical courses, honing the ability to explain diverse statistical topics clearly and accurately.
 - Engaged with high school students at events such as Florence Nightingale Day, translating high-level statistical concepts into accessible and engaging explanations to spark interest in statistics.
- Sep 2022 – **First-year Learning Community Assistant Peer Mentor**, *University of Toronto*
- Apr 2023
- Facilitated the planning and implementation of 12 FLC sessions, coordinating discussions with professors from the Department of Mathematics and library staff.
 - Guided 20+ junior year students in academic matters, fostering skills to overcome challenges and manage transition anxiety to university life.
 - Established a rapport and inclusive community among first-year students, promoting a supportive learning environment.
 - Managed online Quercus page to facilitate remote communication and engagement with first-year students.

Volunteer

- Sep 2022 – **Data Entry Volunteer**, *St. Michael’s Hospital Trauma Research Program and Injury Prevention*
- Dec 2022
- Reviewed eligible medical charts to collect injury / poisoning information
 - Documented medical records into the system